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Project Title	Movie Recommendation System using Matrix Factorization
Week	week 1

WEEK-1 Technical Report

Task Name

Movie Recommendation System using Matrix Factorization

Abstract

A Movie Recommendation System helps users discover movies based on their preferences and viewing history. Matrix Factorization is one of the most effective collaborative filtering techniques used in modern recommendation systems. This project implements Matrix Factorization from scratch using Python, NumPy, and optimization algorithms without relying on specialized recommendation libraries. The system decomposes the user–movie rating matrix into lower-dimensional latent feature matrices to predict unknown ratings and recommend relevant movies. This approach improves recommendation accuracy and provides personalized suggestions for users.

Introduction

With the rapid growth of online streaming platforms, users are faced with thousands of movie choices. Personalized recommendation systems help users find movies that match their interests while increasing user engagement. Matrix Factorization is a widely used collaborative filtering technique that learns hidden relationships between users and movies by analyzing rating patterns. This project focuses on implementing the algorithm from scratch to understand the mathematical concepts behind recommendation systems.

Theory

Matrix Factorization is a collaborative filtering technique that predicts missing ratings by decomposing the user-item rating matrix into two low-dimensional matrices

The model assumes:

- Users and movies can be represented by latent features.
- Similar users have similar preferences.
- Similar movies share common characteristics.
- Unknown ratings can be estimated using latent vectors.

The decomposition is represented as:

$$R \approx P \times Q^T$$

where:

- R = User-Movie Rating Matrix
- P = User Latent Feature Matrix
- Q = Movie Latent Feature Matrix

The predicted rating is calculated as:

$$\hat{R} = P \times Q^T$$

These parameters collectively describe purchasing frequency and customer retention.

Objectives

The objectives of this project are:

- Develop Matrix Factorization from scratch.
- Learn latent features of users and movies.
- Predict unknown movie ratings.
- Recommend personalized movies.
- Minimize prediction error using Gradient Descent.
- Evaluate recommendation accuracy.

Methodology

The implementation consists of the following stages:

Step 1: Data Collection

Collect user ratings for different movies.

Step 2: Data Preprocessing

- Remove duplicate records.
- Handle missing ratings.
- Create the User-Movie Rating Matrix.

Step 3: Model Development

Initialize user and movie latent matrices with random values.

Step 4: Training

Update latent matrices using Gradient Descent until prediction error is minimized.

Step 5: Recommendation

Predict unrated movies and recommend those with the highest predicted ratings

Dataset

The dataset contains:

- User ID
- Movie ID
- Movie Title
- Rating (1–5)
- Timestamp

The model can be trained using publicly available datasets such as MovieLens.

Mathematical Likelihood

Given:

$$R \in \mathbb{R}^{(m \times n)}$$

Find:

$$P \in \mathbb{R}^{(m \times k)}$$

$$Q \in \mathbb{R}^{(n \times k)}$$

such that:

$$R \approx P \times Q^T$$

The optimization objective is:

Minimize

$$[J = \sum_{(i,j) \in K} (R_{ij} - P_i Q_j^T)^2 + \lambda (|P|^2 + |Q|^2)]$$

where:

- K = Set of known ratings
- λ = Regularization parameter

Complexity Analysis

- Training Complexity: $O(K \times \text{Iterations})$
- Prediction Complexity: $O(m \times n \times k)$
- Memory Complexity: $O((m+n) \times k)$

where:

- m = Number of users
- n = Number of movies
- k = Number of latent features

Results

The implemented recommendation system successfully:

- Learns hidden user preferences
- Predicts missing ratings.

- Generates personalized movie recommendations.
 - Reduces prediction error through optimization.
 - Performs efficiently on large datasets.
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Applications

- Movie Recommendation Platforms
 - OTT Streaming Services
 - E-commerce Product Recommendation
 - Online Learning Course Recommendation
 - News Recommendation Systems
 - Music Recommendation Systems
 - Social Media Content Recommendation
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Advantages

- High recommendation accuracy.
- Handles sparse datasets effectively.
- Learns hidden user interests automatically.
- Learns hidden user interests automatically.
- Scalable for large datasets.
- Improves user experience through personalization.

Conclusion

This project presents a Movie Recommendation System using Matrix Factorization implemented from scratch. By decomposing the user-item rating matrix into latent feature representations, the system effectively predicts user preferences and recommends relevant movies. Matrix Factorization provides accurate, scalable, and personalized recommendations, making it one of the most widely used techniques in modern recommendation engines such as streaming platforms and e-commerce applications.